

VEL'MO, Russian Federation

WMO: 239820

Lat: 61.0168N

Lon: 93.4281E

Elev: 400

StdP: 14.48

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-53.6	-47.0	-57.7	0.2	-53.4	-51.1	0.3	-46.8	15.8	-2.0	14.2	-6.5	0.5	130	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	84.5	66.1	80.9	65.3	77.2	63.9	69.3	80.0	67.2	77.6	65.0	74.5	5.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	95.6	74.7	63.3	88.2	71.8	61.1	81.8	69.9	33.8	79.8	32.0	78.1	30.3	74.1	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.3	11.6	DB	-56.0	88.8	7.3	3.5	-61.3	91.3	-65.6	93.4	-69.7	95.3	-74.9	97.9	(4)
(5)			WB	-56.9	73.2	8.7	2.4	-63.2	74.9	-68.3	76.3	-73.2	77.7	-79.5	79.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.1	-17.1	-7.1	13.2	28.3	40.7	56.9	63.7	56.6	43.7	26.5	0.8	-7.2	(6)
(7)		DBStd	29.70	17.59	16.04	14.22	9.18	7.79	9.12	5.95	6.51	7.67	10.70	16.68	19.35	(7)
(8)		HDD50	10030	2081	1598	1141	651	308	41	0	17	218	728	1475	1772	(8)
(9)		HDD65	14652	2546	2018	1606	1101	753	268	97	270	639	1192	1925	2237	(9)
(10)		CDD50	943	0	0	0	0	20	247	426	221	29	0	0	0	(10)
(11)		CDD65	91	0	0	0	0	0	24	58	9	0	0	0	0	(11)
(12)		CDH74	1525	0	0	0	0	17	594	770	137	7	0	0	0	(12)
(13)		CDH80	393	0	0	0	0	3	169	202	19	0	0	0	0	(13)
(14)	Wind	WSAvg	4.1	2.9	3.0	4.7	5.8	5.5	4.5	4.2	3.5	3.9	4.1	3.6	3.2	(14)
(15)	Precipitation	PrecAvg	22.7	1.2	0.9	1.0	1.2	1.9	2.5	2.0	3.1	3.0	2.2	2.1	1.5	(15)
(16)		PrecMax	27.1	2.6	1.9	2.4	3.1	4.1	4.6	4.6	5.6	5.6	4.6	3.4	3.2	(16)
(17)		PrecMin	17.6	0.5	0.0	0.1	0.5	0.7	0.5	0.5	0.5	1.4	0.9	0.8	0.6	(17)
(18)		PrecStd	2.9	0.5	0.4	0.5	0.6	0.7	1.0	0.9	1.5	1.1	0.8	0.6	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.7	30.1	45.4	55.7	74.7	89.0	88.8	82.6	74.0	53.2	33.7	28.2	(19)
(20)			MCWB	20.3	28.0	36.3	44.4	55.9	69.0	66.9	66.2	62.0	44.3	31.7	27.2	(20)
(21)		2%	DB	17.0	25.8	39.9	50.2	68.1	84.9	85.1	77.4	65.8	46.2	30.5	25.3	(21)
(22)			MCWB	15.6	24.2	32.7	39.9	53.2	65.8	66.5	64.7	56.5	41.1	29.3	24.1	(22)
(23)		5%	DB	12.0	21.5	36.1	45.8	61.6	81.0	81.8	74.0	60.3	42.7	27.4	22.4	(23)
(24)			MCWB	11.1	19.8	31.6	37.3	48.9	64.2	66.8	62.9	53.2	39.4	26.3	21.3	(24)
(25)		10%	DB	7.8	16.3	33.0	41.9	55.5	76.7	78.2	70.3	56.1	39.6	25.1	19.0	(25)
(26)			MCWB	7.1	15.1	29.6	35.4	46.2	62.8	64.7	61.0	50.8	37.0	24.0	18.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	27.9	37.2	44.6	57.7	71.8	72.2	70.1	63.3	45.9	32.2	27.0	(27)
(28)			MDCB	21.5	30.0	44.1	54.0	72.2	83.2	81.8	77.7	71.7	50.5	33.1	27.7	(28)
(29)		2%	WB	15.5	24.3	33.8	41.3	54.5	68.4	70.0	66.6	58.3	42.1	29.2	24.3	(29)
(30)			MDCB	16.7	25.7	38.7	48.0	65.3	82.4	80.3	74.9	63.3	45.7	30.2	25.4	(30)
(31)		5%	WB	11.1	20.0	31.7	38.3	50.9	65.8	68.1	64.2	54.2	39.7	26.4	21.4	(31)
(32)			MDCB	11.9	21.4	35.5	44.3	60.6	77.8	78.3	72.3	59.0	42.3	27.2	22.4	(32)
(33)		10%	WB	7.1	15.2	29.6	36.0	46.8	63.1	66.1	61.8	51.1	36.9	23.8	17.9	(33)
(34)			MDCB	7.8	16.3	33.0	41.0	54.4	74.7	75.6	68.2	55.5	39.6	24.9	19.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.4	22.3	24.6	19.7	19.9	24.5	23.6	20.9	15.6	11.8	16.9	16.4	(35)
(36)			MCDBR	18.8	17.9	22.2	24.7	32.1	33.9	30.0	28.6	25.3	15.4	13.4	12.5	(36)
(37)			MCWBR	17.9	16.6	16.7	15.8	18.1	17.5	14.9	16.9	15.8	10.7	12.9	11.8	(37)
(38)		5% WB	MCDBR	18.9	17.4	19.9	21.1	28.8	30.7	25.7	24.0	22.4	14.2	12.8	12.6	(38)
(39)			MCWBR	18.0	16.1	15.3	14.6	17.6	17.5	14.7	15.2	14.9	10.6	12.6	12.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.260	0.297	0.328	0.359	0.374	0.389	0.439	0.410	0.348	0.314	0.253	0.228	(40)	
(41)		taud	1.940	1.992	2.012	2.050	2.262	2.377	2.220	2.276	2.421	2.260	1.996	1.884	(41)	
(42)		Ebn at Noon	133	204	243	260	268	266	246	244	243	204	136	90	(42)	
(43)		Edh at Noon	15	26	36	43	38	34	39	34	25	20	13	9	(43)	
(44)	All-Sky Solar Radiation	RadAvg	122	410	833	1293	1552	1748	1574	1127	673	296	139	60	(44)	
(45)		RadStd	16	33	88	138	140	197	141	92	105	49	23	11	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	+5.54	+5.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page